Fields and Waves I

Lecture 7

Impedance Matching with Transmission Line Stubs

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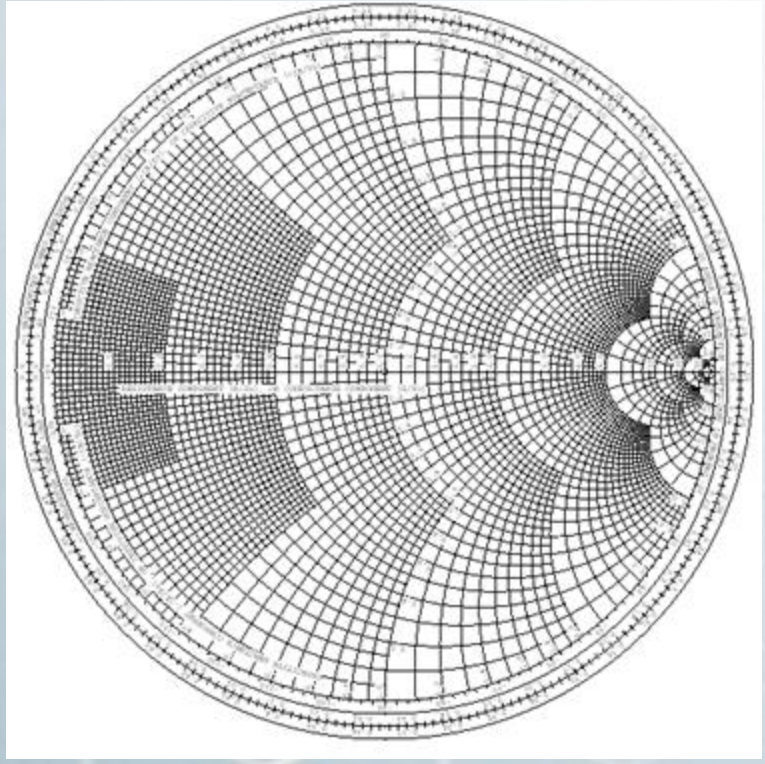
Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Single Stub Matching
- Project 1 Introduction

Review

- The Smith Chart allows for the graphical solution of the transmission line equation for *Z.*
- The Chart gives a direct conversion between Γ and *Z*



Review

solutions of constant x_L

solutions of constant r_L

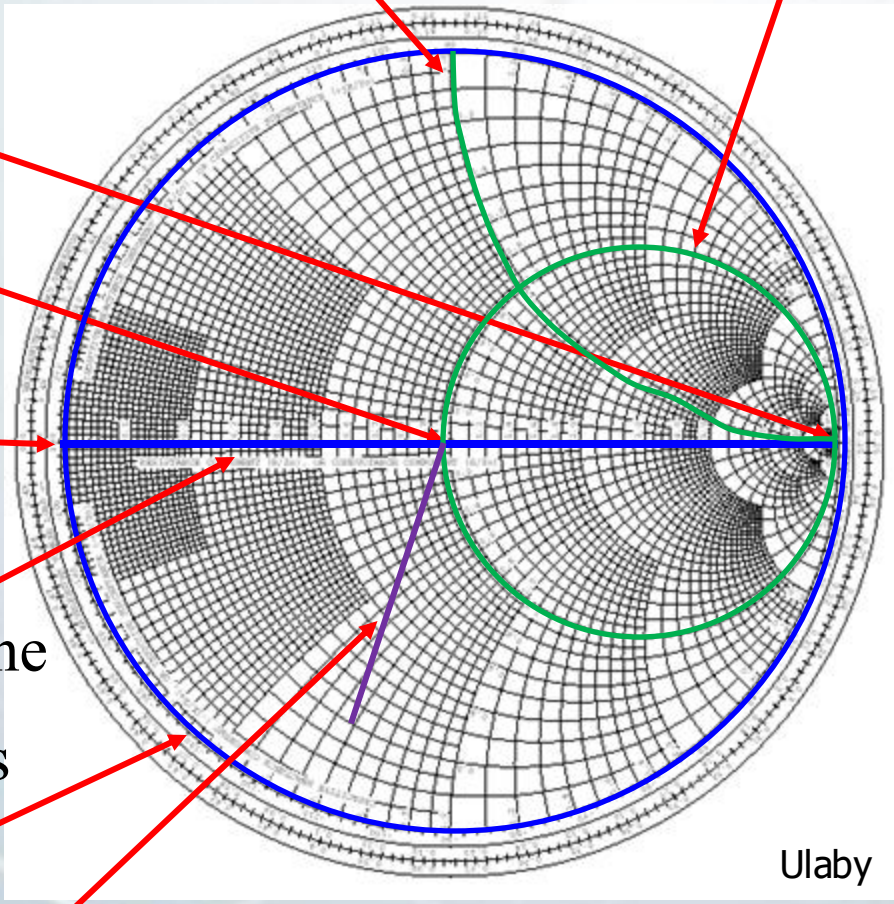
open circuit point (infinite impedance)

unity impedance z =1 (match point)

short circuit point (zero impedance)

purely real impedances are along the horizontal center line

purely imaginary impedances are along the periphery



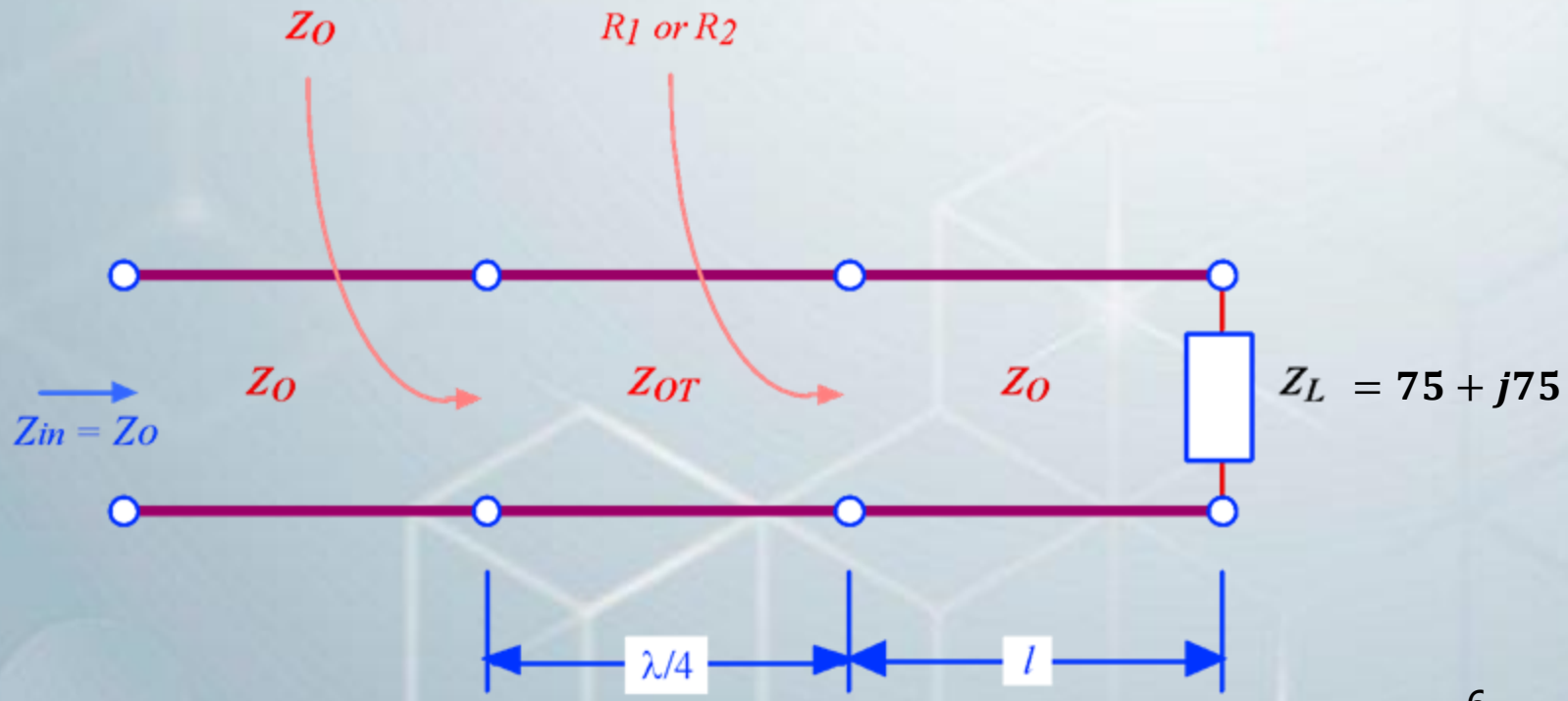
length = $|\Gamma|$

angle with respect to x-axis (counterclockwise) = $\angle\Gamma$

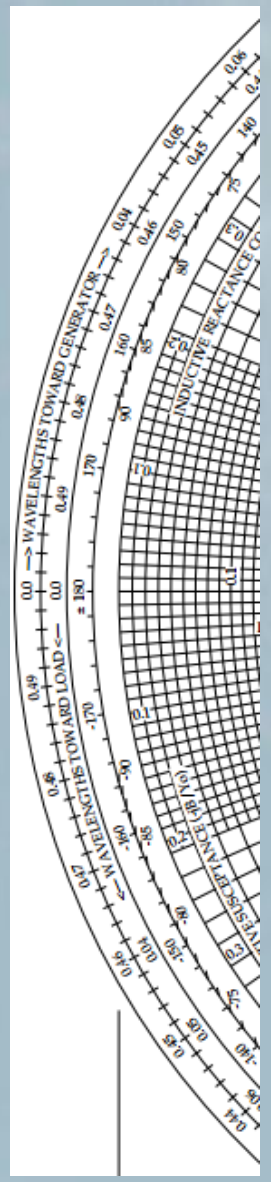
Review

We can match a load to a transmission line using:

1. A specific length of t-line l attached to the load that makes the input impedance looking down the t-line **REAL**
2. A $\lambda/4$ transformer attached to the input of the t-line in (1) to make the input impedance looking down the series combination equal to the impedance we want to match (Z_0 here)



Review

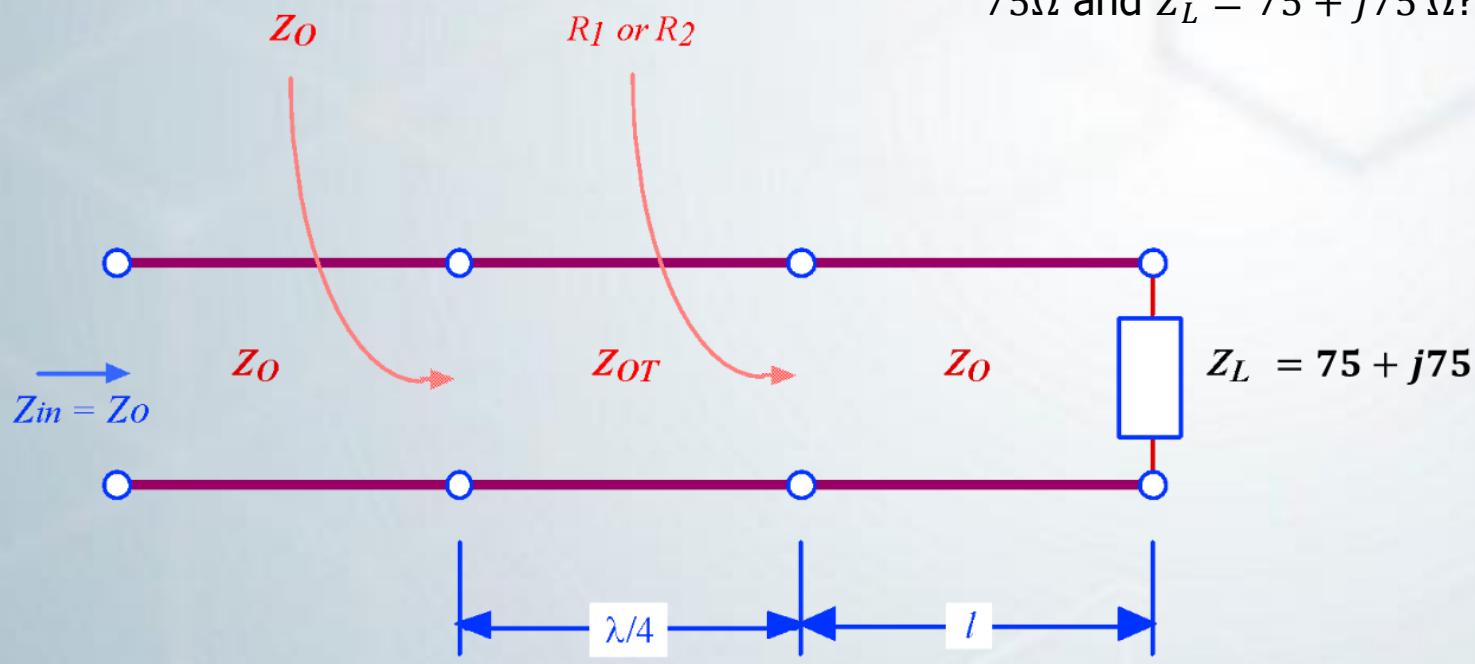


- We can use the Smith chart to find the length of line L required to give a certain input impedance.
- We **start at the normalized load impedance**, travel along a circle **towards the generator**, and stop at the input impedance we want to provide
- Where we stop is the **input impedance of the t-line of length L**. But how do we read off this length?
- There are two scales on the periphery (in wavelengths)
  - 1 **towards generator** (clockwise)
  - 1 towards load (counterclockwise)
- Read off the distance (in wavelengths) you traveled from the initial impedance to the final impedance. This is the line length L.
- Note that once around the whole chart is a total length of λ/2

UQ

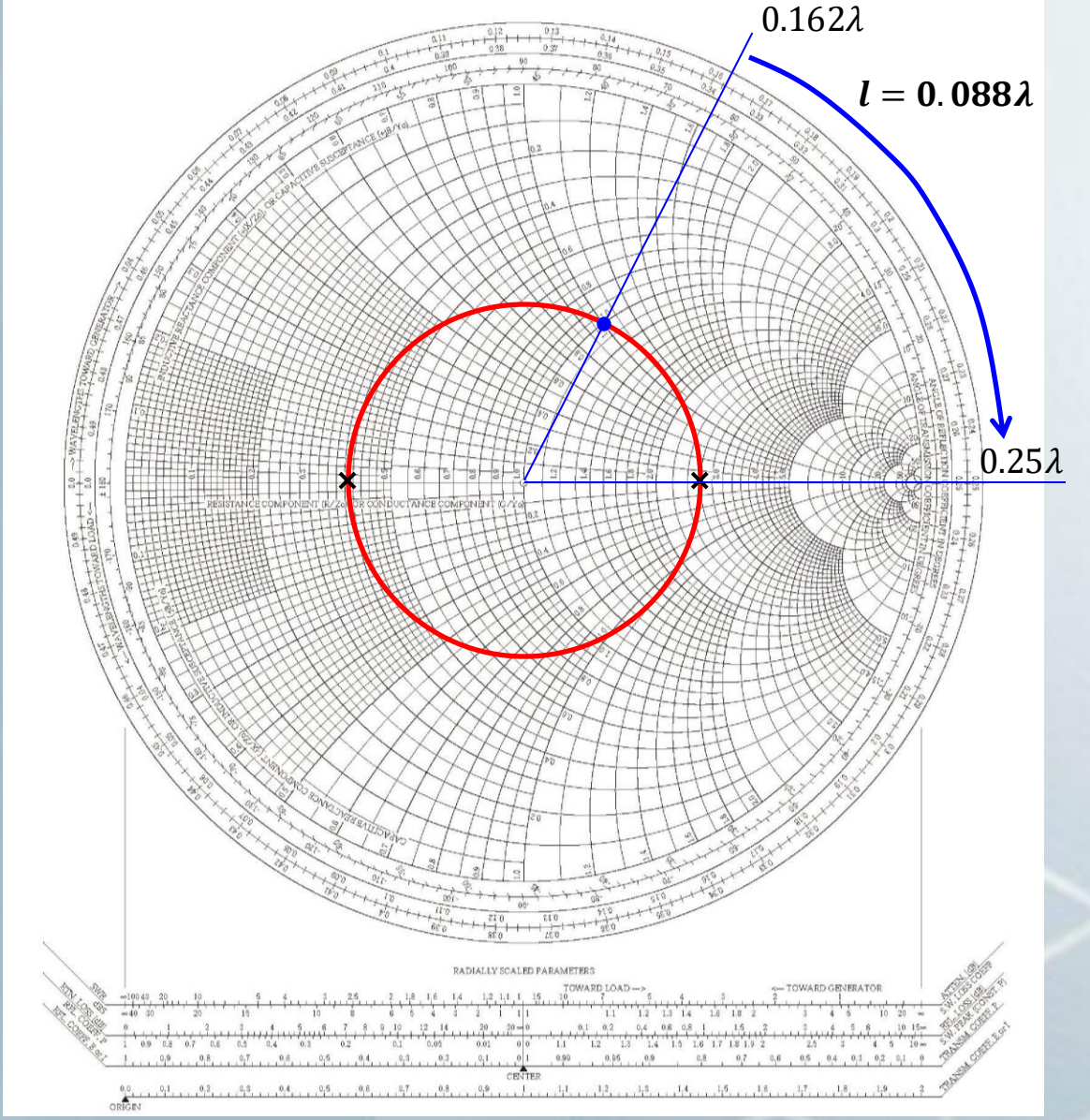
Review

- What length of line l presents a **real** input impedance, when $Z_0 = 75\Omega$ and $Z_L = 75 + j75\ \Omega$?



1. Plot the normalized load impedance z_L on the Smith chart
2. Draw a circle of constant radius through the plotted impedance (centered at origin of Smith chart) – VSWR circle
3. Find where the circle intersects the x-axis of the chart – these are all the possible real input impedances
4. Choose a real input impedance and read off the required l

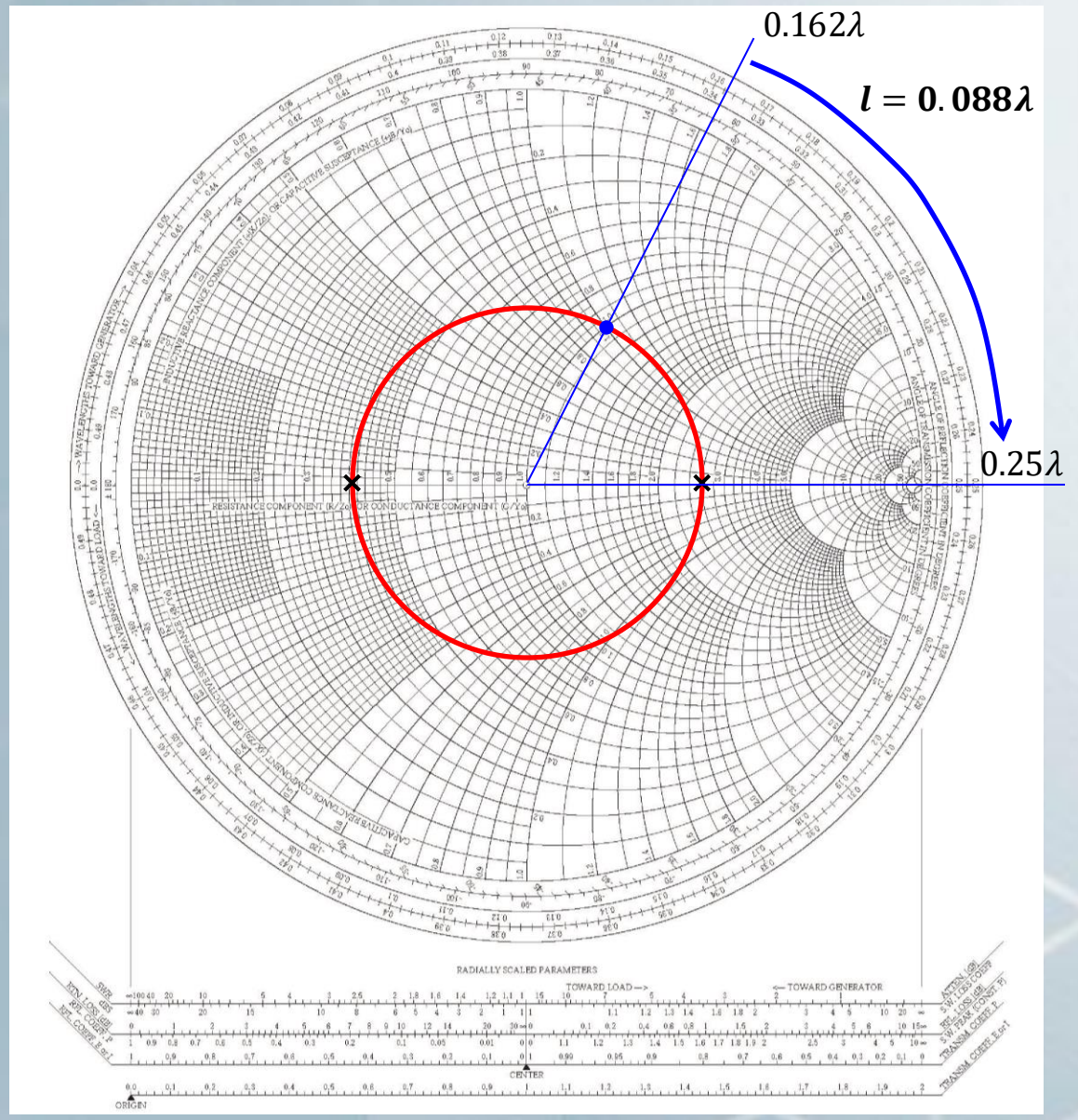
Review



- Read the normalized input resistance from the Smith chart and convert it to its actual input resistance.

$$r = 2.6 \rightarrow$$
$$R = 2.6 \cdot 75\Omega = 195\Omega$$

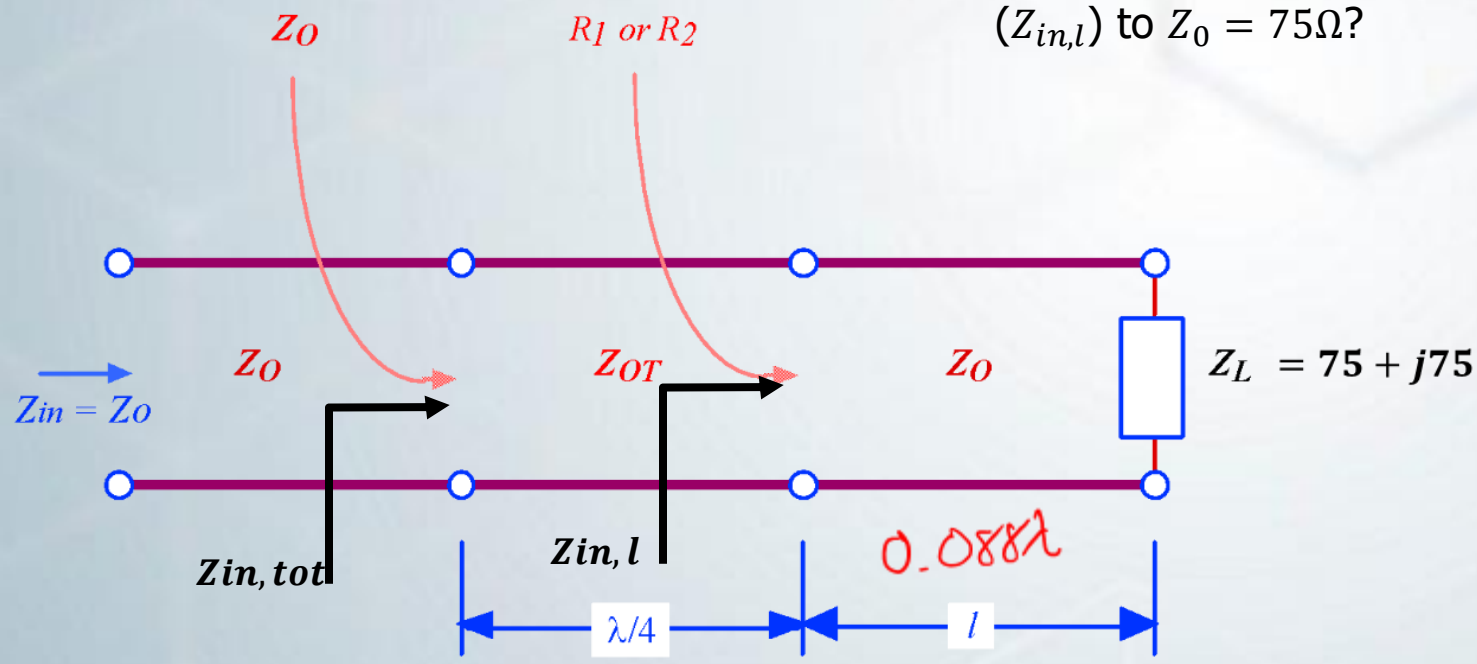
Review



- Choose a real input impedance and read off the required l
  - (1) Draw a line from the origin through the load impedance to the edge of the chart. Read off the number on the "Wavelengths toward generator" scale.
  - (2) Do the same for your chosen real input impedance.
  - Calculate the total distance required to travel from (1) to (2)

Review

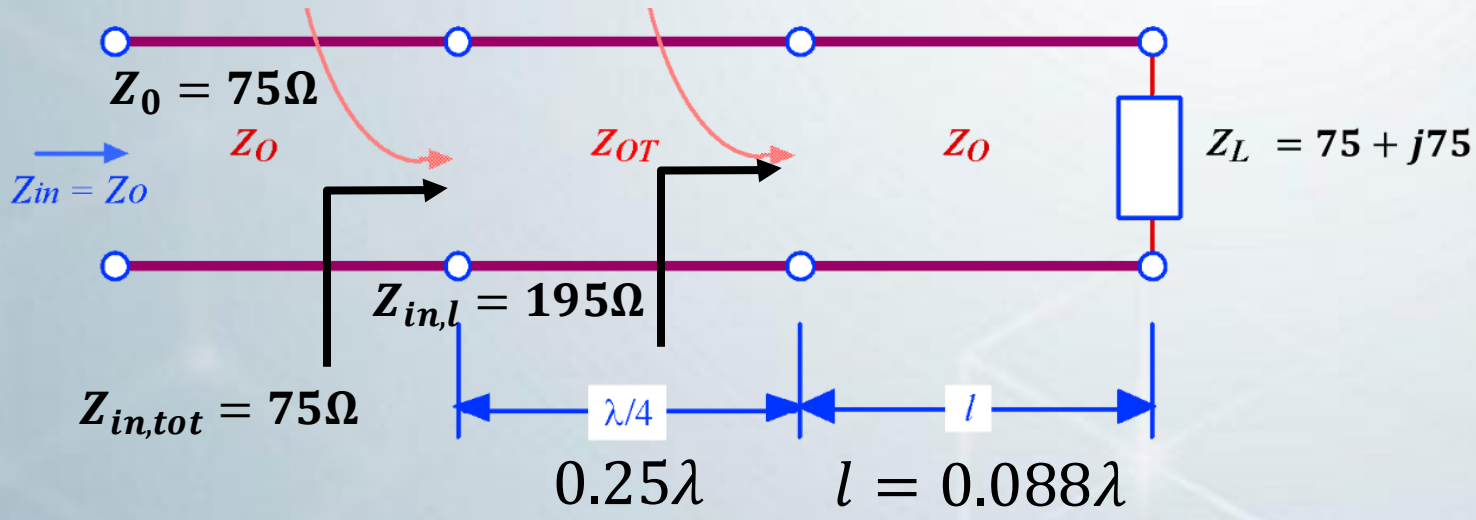
- What characteristic impedance Z_{OT} will match the input impedance of the line of length l ($Z_{in,l}$) to $Z_0 = 75\Omega$?



1. Read the normalized input resistance from the Smith chart and convert it to its actual input resistance.
2. Calculate Z_{OT} that is required to match $Z_{in,l}$ to $Z_0 = 75\Omega$

$$Z_{in,tot} = \frac{Z_{OT}^2}{Z_L} \rightarrow Z_{OT} = \sqrt{Z_{in,tot} \cdot Z_L} = \sqrt{75\Omega \cdot 195\Omega} = \mathbf{120.9\Omega}$$

Review



We have matched the feedline with $Z_0 = 75\Omega$ to a load impedance $Z_L = 75 + j75\Omega$!

Review

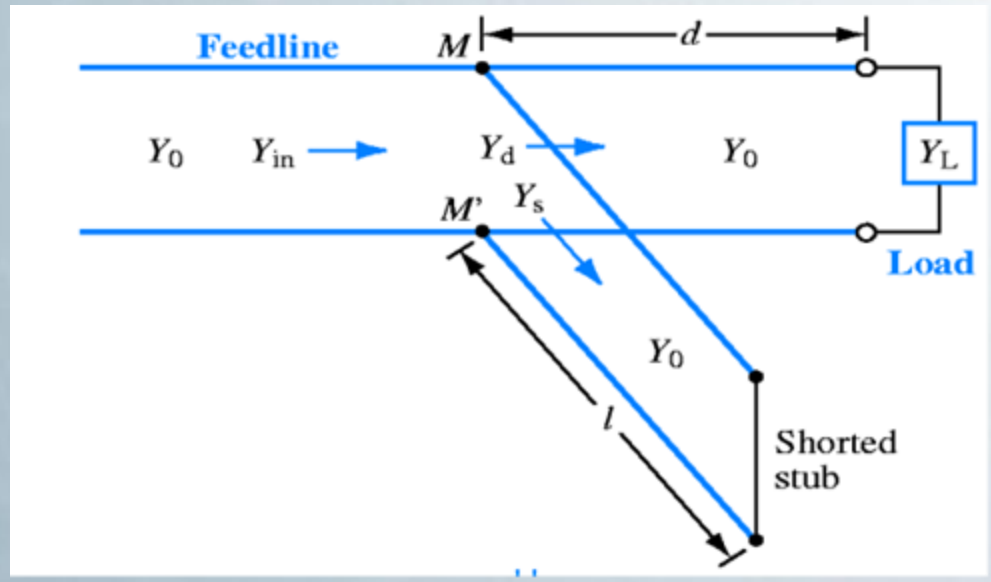
- Let's say you only have one type of cable and can't design a quarter-wavelength transformer with arbitrary Z_0. How do you match then?
- You could add inductors or capacitors to the network in order to match impedance. But in practice, we can't always get the L and C values that we need to match properly.
- Another approach is to use a parallel stub of cable: this is called

Single-stub matching

Single-Stub Impedance Matching

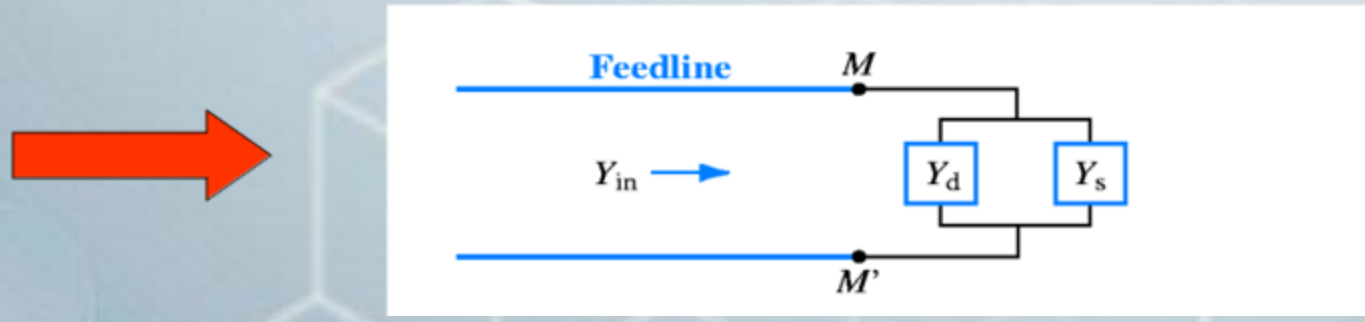
Single Stub Matching

- Instead of using a t-line to provide a real input impedance, then using a quarter wave transformer, we are going to cancel out the imaginary part of Z_L by placing a short length of t-line (a stub) in parallel with the load



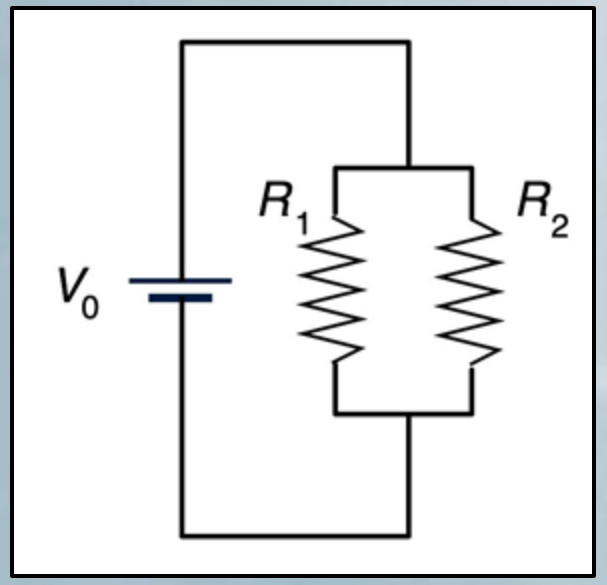
But how does this work?

Ulaby



Single Stub Matching

- Reminder: admittances are very useful when working with parallel circuit elements because parallel admittances simply add together.
- If we want to find $Z_{eq} = Z_L || Z_{in}$,we have to use the following equation:



$$Z_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{Y_{eq}}$$

$R_1 \sim Y_1$, $R_2 \sim Y_2$

- However, if we consider their admittances, they just add:

$$Y_{eq} = \frac{1}{R_1} + \frac{1}{R_2} = Y_1 + Y_2$$

Single Stub Matching

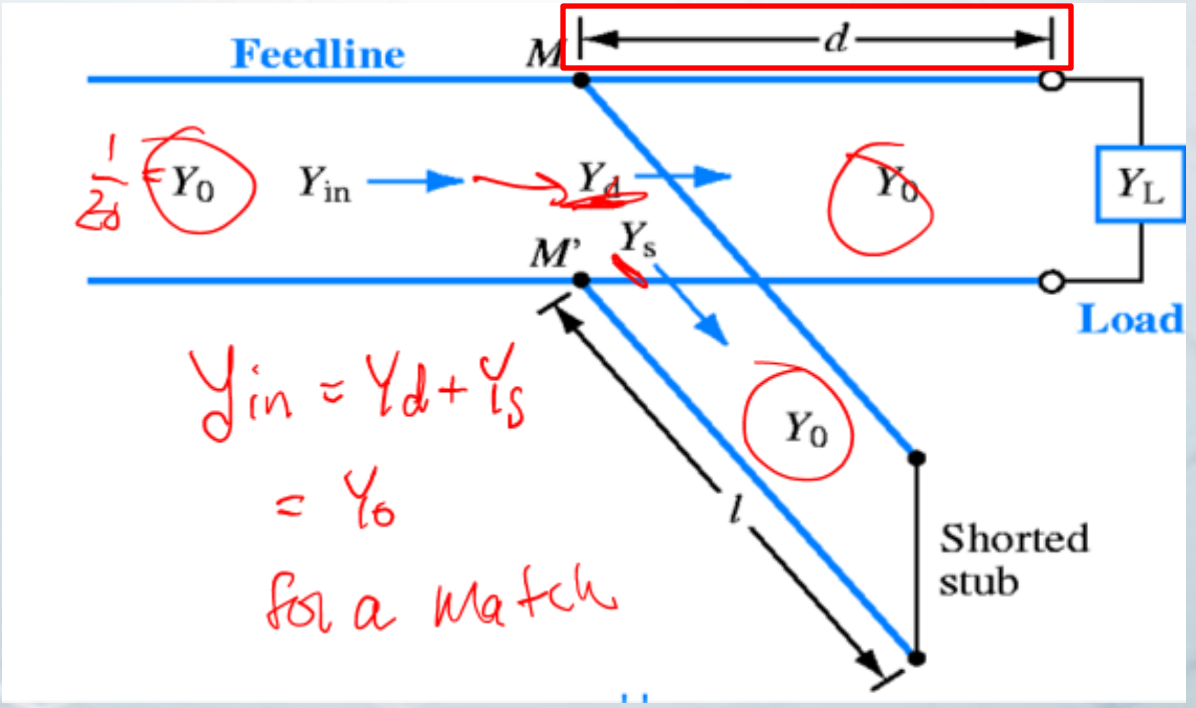
- The idea: we first convert our load impedance to a **load admittance Y_L**
- Next, we find a length of t-line d that gives an input admittance of the form:

$$Y_d = (Y_0 + jB)$$

This means the real part of our input admittance, Y_d, is matched to the characteristic admittance of our feedline: $Y_0 = \frac{1}{Z_0}$

- To complete the match, we just need to find a length of t-line (a stub) with input admittance:

$$Y_s = -jB$$



- If we place the stub in parallel with the first t-line of length d, the admittances add, giving a total input admittance: $Y_d + Y_s = Y_0$ **This is a match!**

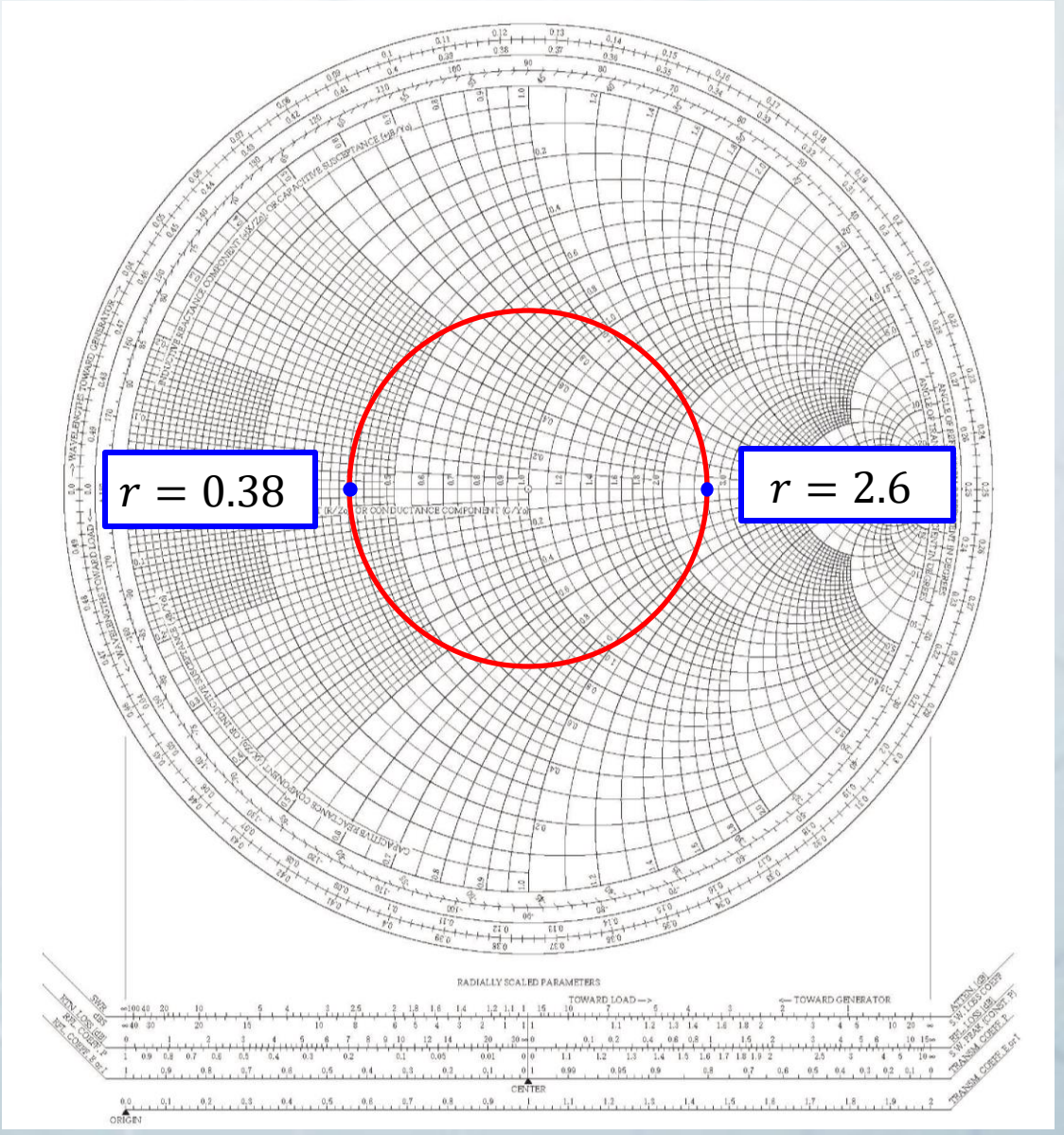
Single Stub Matching

- First, we need to learn how to use the Smith chart to find **admittances** instead of impedances

- What do you notice about the relationship between these two real impedances' magnitudes and positions on the Smith Chart?

$\frac{1}{0.38} = 2.6$ ← right-hand sol'n

↑ left hand solution



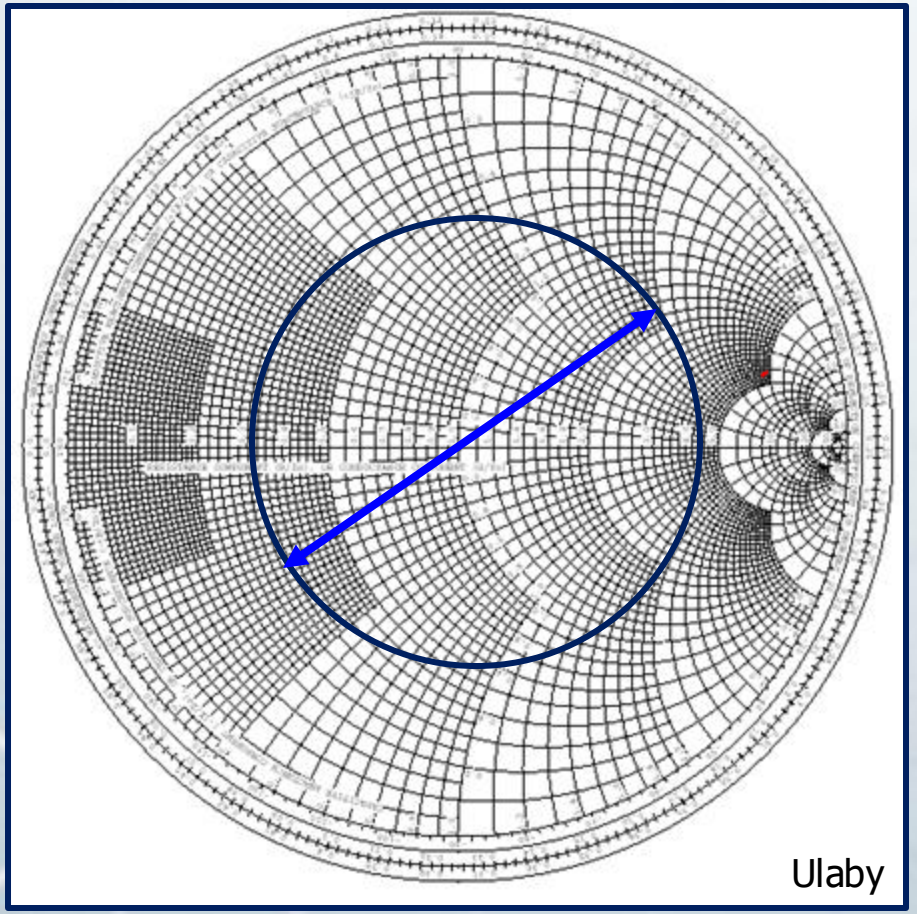
Single Stub Matching

- Any point reflected through the center point converts impedance (Z) to admittance (Y) and vice versa.

Top Half: inductive reactance
or
capacitive susceptance

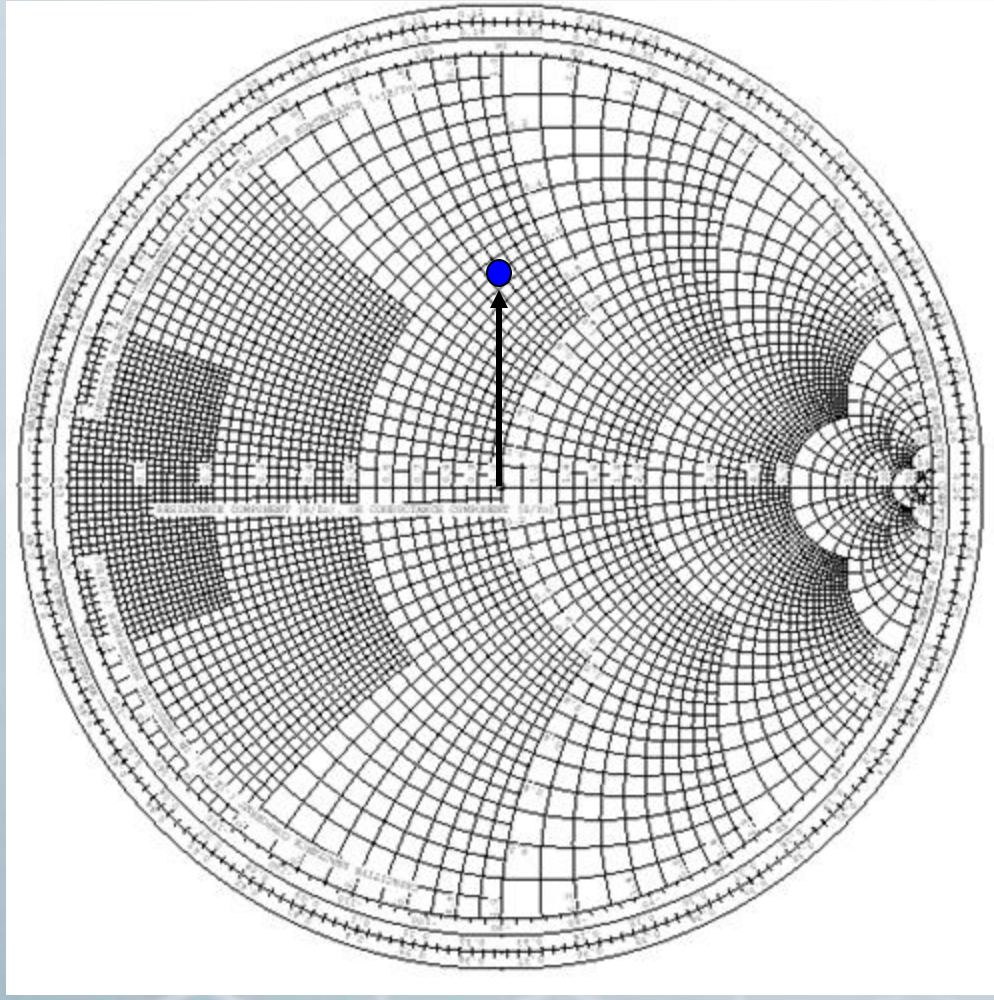
usually marked on charts

Bottom Half: capacitive reactance
or
inductive susceptance



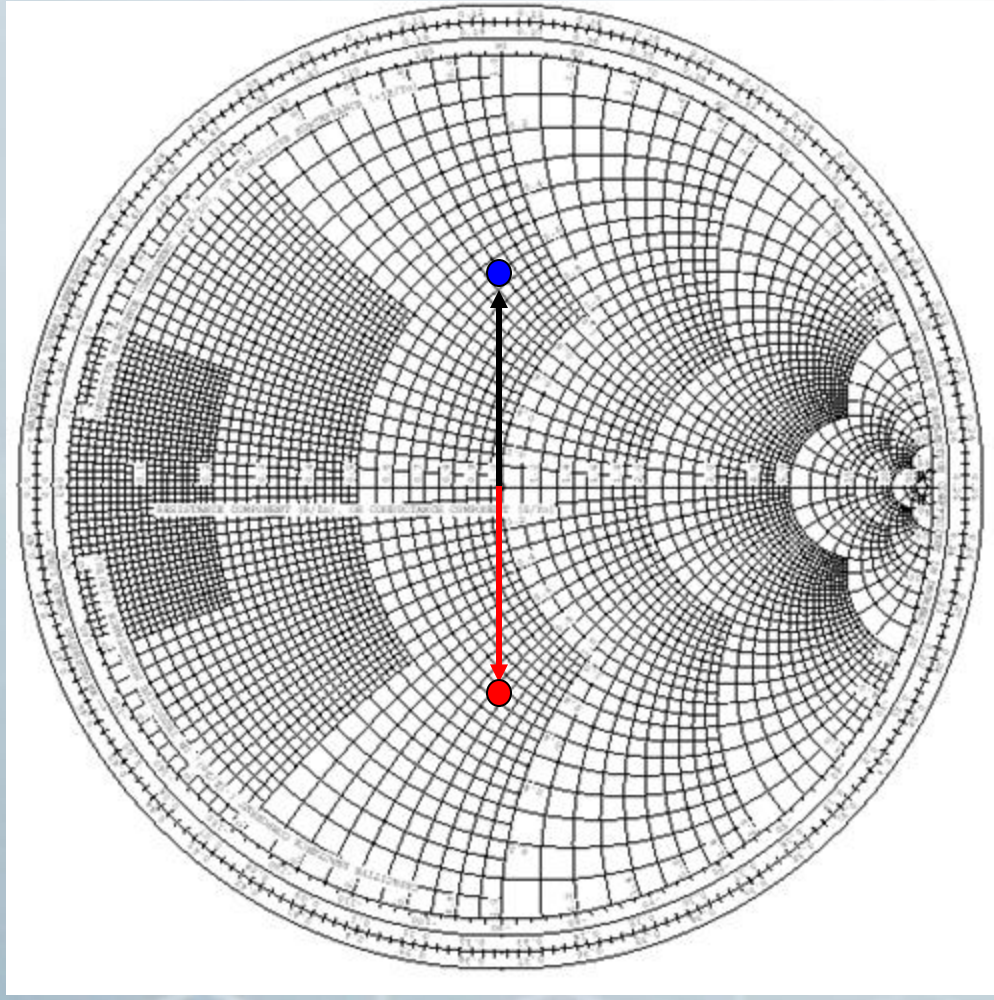
Single Stub Matching

- Consider this example. The blue dot represents the input impedance at the end of a λ/8 transmission line. How do we find the input admittance?



Single Stub Matching

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Single Stub Matching

- Load of 50 + j50 Ohms on 50 Ohm Transmission Line
  - What is the normalized impedance? Plot it on the Smith chart

$$z_e = 1 + j$$

  - What is the normalized admittance? Plot it on the Smith chart

$$y_e = \frac{1}{1+j} = 0.5 - j0.5$$

- Note! Once we cross the center of the chart to enter the "admittance domain" all numbers on the Smith chart are to be considered admittances
  - What is the admittance of a short circuit? Where is it on the Smith chart?

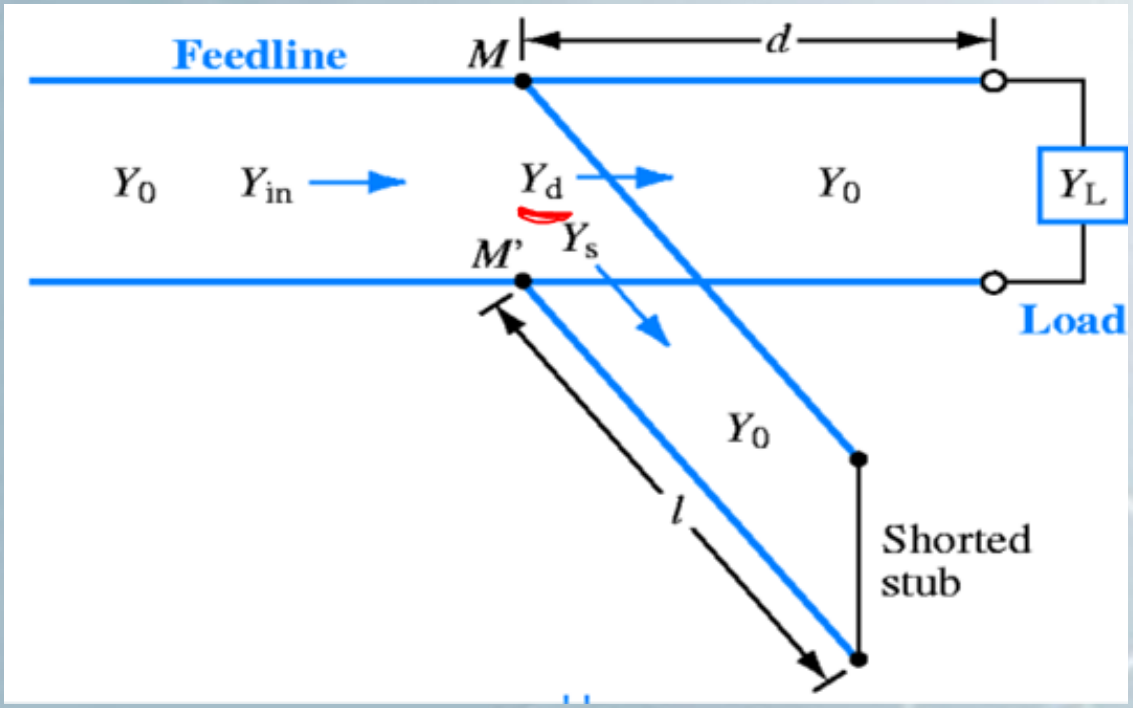
$y_{sc} = \infty$ right-most point

  - What is the admittance of an open circuit? Where is it on the Smith chart?

$y_{oc} = 0$ left-most point

Single Stub Matching

- Now that we know how to find admittances on the Smith chart, let's return to single stub matching
- If we normalize our admittances from a couple of slides ago (divide them by the characteristic admittance Y_0), we have:



$$Y_d = (Y_0 + jB) \rightarrow y_d = 1 + jb$$

$$Y_s = -jB \rightarrow y_s = -jb$$

$$Y_d + Y_s = Y_0 \rightarrow y_{in} = 1$$

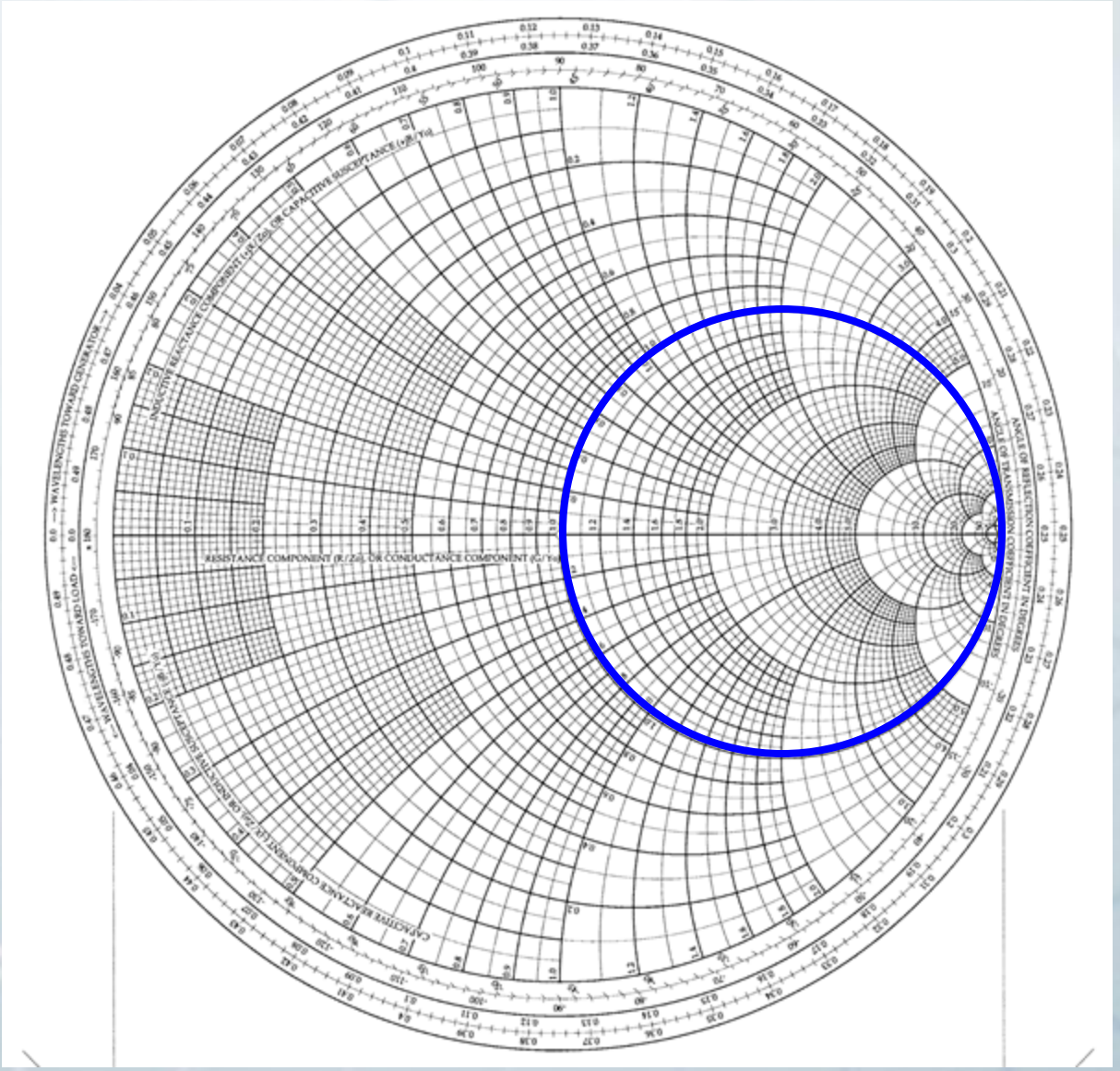
Match!

Our task is then to:

1. Find the length d that gives $y_d = 1 + jb$
2. Find the length l that gives $y_s = -jb$

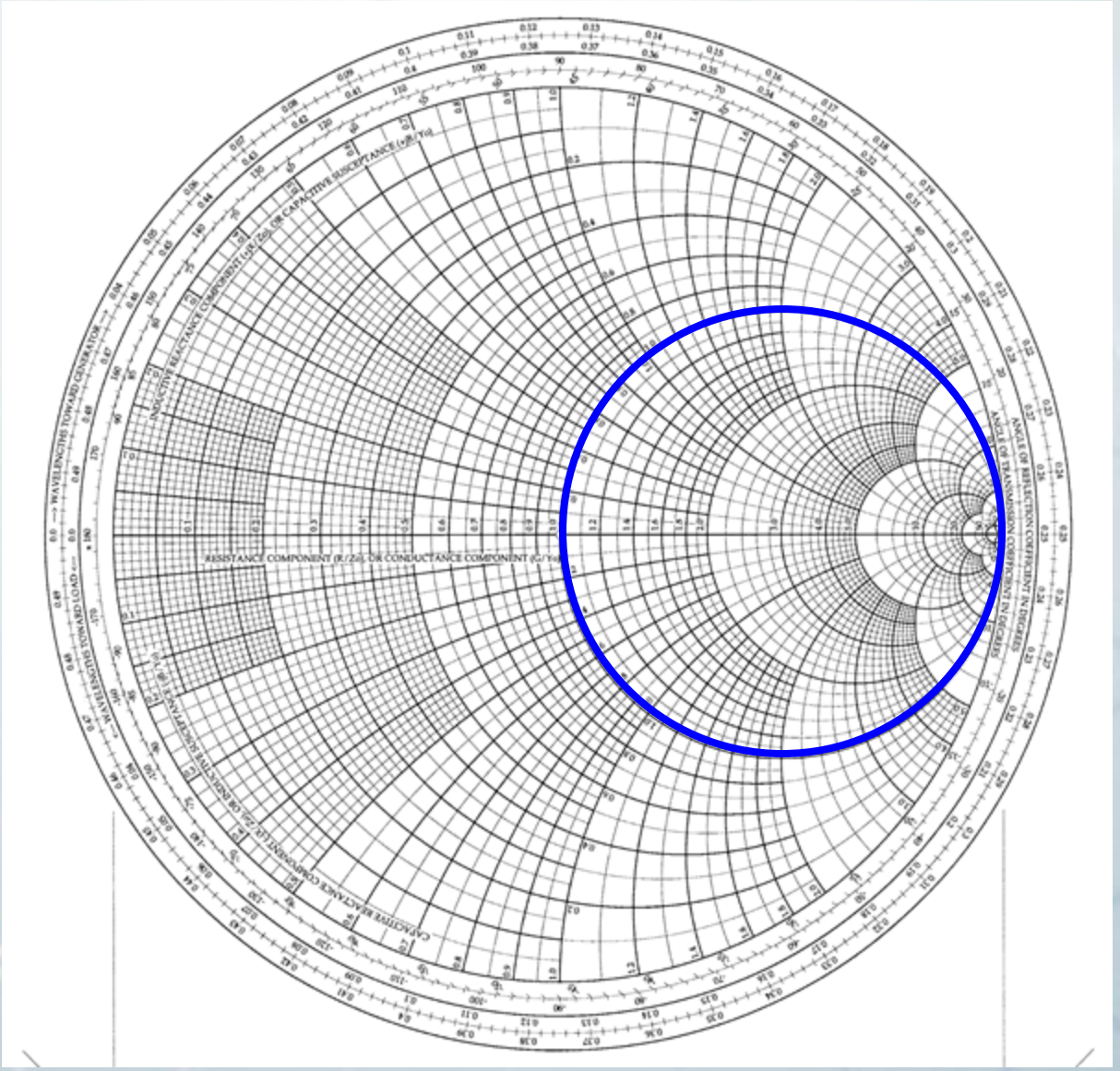
Single Stub Matching

- Consider this circle on the Smith chart. Its leftmost edge touches the origin of the chart and its rightmost edge touches the rightmost edge of the chart.
- The impedances that land on this circle are all those impedances of form **1+jx** (i.e. Z_0 plus some reactive impedance.)
- We call this the **matching circle**, since R_L is matched to Z_0 at every point on the circle.



Single Stub Matching

- It works for admittances as well.
- On this circle, if $z = r + jx$, then $r = 1$ and $Z_0 = Re\{Z_L\}$ (matched)
- Likewise, $\frac{1}{Z_0} = 1/Re\{Z_L\}$, which is equivalent to $Y_0 = Re\{Y_L\}$, meaning the real parts of Z and Y are matched to the load on the matching circle
- That leaves the reactance (x) and susceptance (b) as the only terms left to match. On this circle, admittances are of the form $y = 1 + jb$.



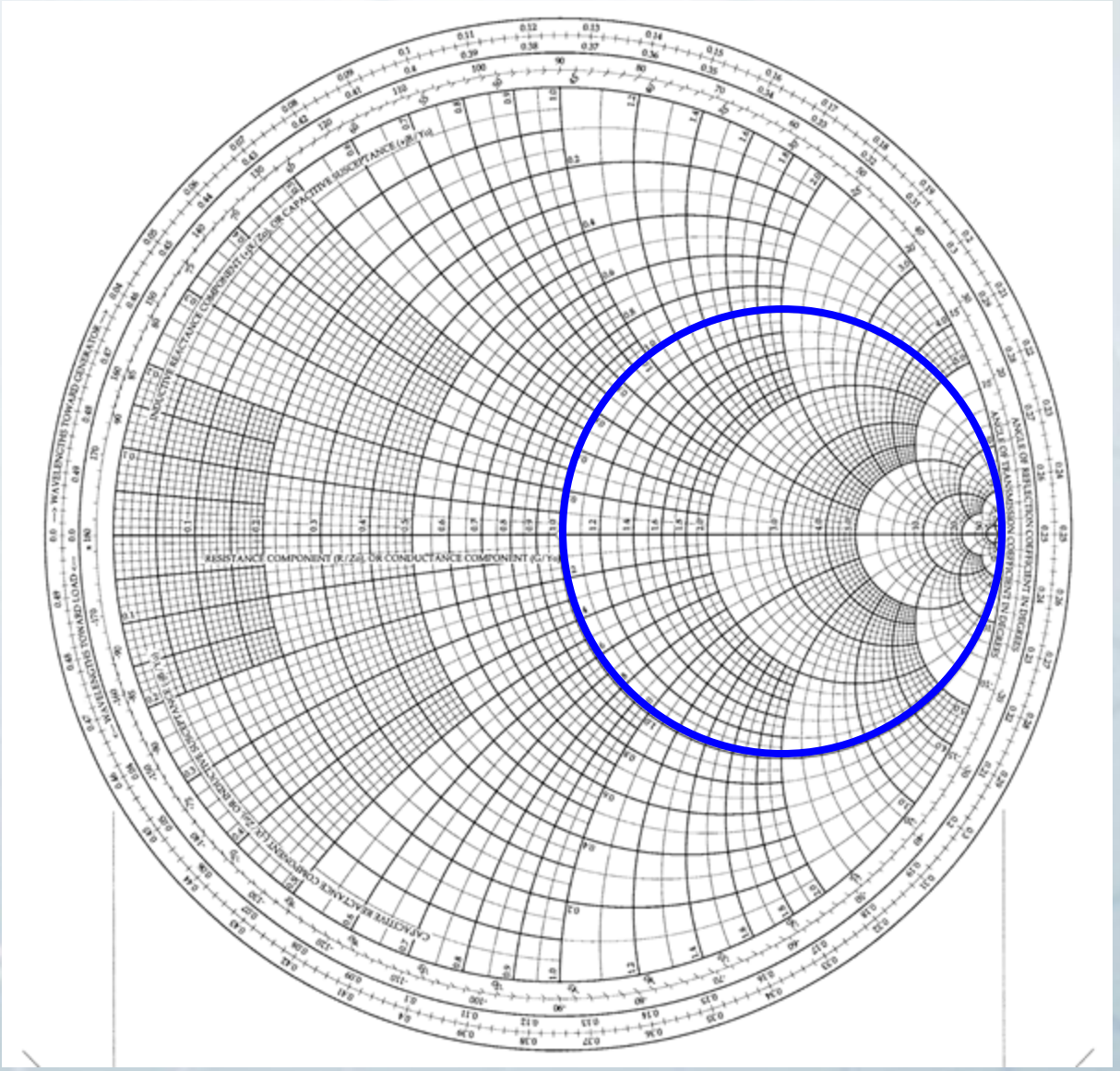
We can cancel jb with a stub.

Single Stub Matching

- This means that this circle contains all admittances that can be matched to Y_0 $(y = 1)$ by adding a stub:

$$y_{in} = y_d + y_s$$
$$= (1 + jb) + (-jb)$$
$$= 1$$

- If we can find a length of t-line that will put us on the matching circle, we just need to add a length of t-line (a stub) in parallel with the input that cancels out jb



Single Stub Matching

- We already know how to find a length of t-line that provides us with a certain input impedance, given the load impedance.
- The process is the same for admittances:
  1. Plot the normalized load admittance y_l
  2. Draw the VSWR circle
  3. Travel along the VSWR circle from the load admittance towards the generator to a location on the matching circle ($y = 1 + jb$)
  4. Read off the input admittance y_d and line length d from the Smith chart

What about this "stub"? How do we find a length of t-line that gives us an input admittance $y_s = -jb$?

Single Stub Matching

- Recall that at any given frequency, by choosing the length l, we can cause a length of t-line with either a short circuit or an open circuit at the end to have any **reactive** input impedance we want:

$$Z_{in}^{sc} = jZ_0 \tan(\beta l)$$

$$Z_{in}^{oc} = -jZ_0 \cot(\beta l)$$

- The same is also true of admittance. We can use a short- or open-circuited stub with a specific length to add an arbitrary **susceptance** (imaginary admittance) to a network:

$$Y_{in}^{sc} = -jY_0 \cot(\beta l)$$

$$Y_{in}^{oc} = jY_0 \tan(\beta l)$$

Single Stub Matching

- We now have all the tools we need to perform matching with a single stub. Here is the procedure:

1. Convert the load impedance Z_L to an equivalent normalized admittance $y_L = 1/z_L$.
2. Use a length of line of characteristic impedance Z_o to transform y_L to $\boldsymbol{y_d = 1 + jb}$.
   Note: *the term "1" means we have matched the real part of* Z_L to Z_0
3. Add a stub in parallel with y_d which has an input admittance $\boldsymbol{y_s = -jb}$.
4. Therefore, the total admittance at the input to the parallel combination of y_d and y_s is:

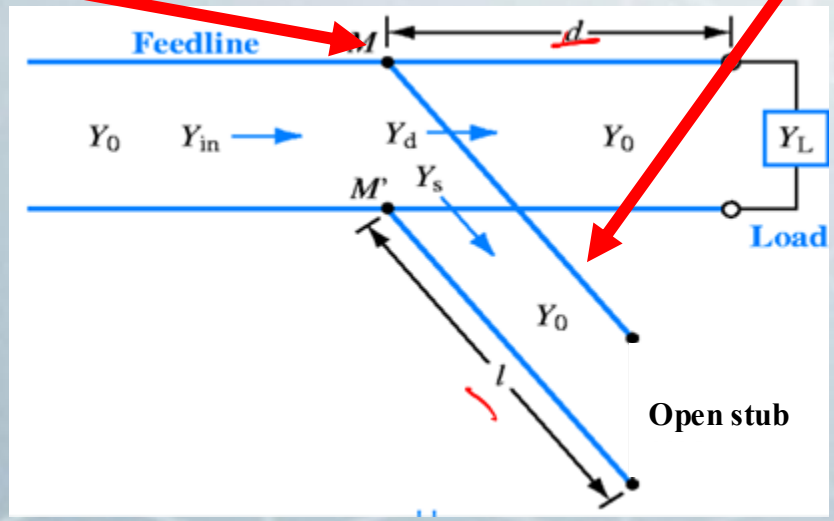
$$\boldsymbol{y_{in} = y_d + y_s = (1 + jb) + (-jb) = 1 \rightarrow Y_{in} = Y_0} \text{ and } \boldsymbol{Z_{in} = Z_0}$$

Match!

Single Stub Matching

Let's do an example:

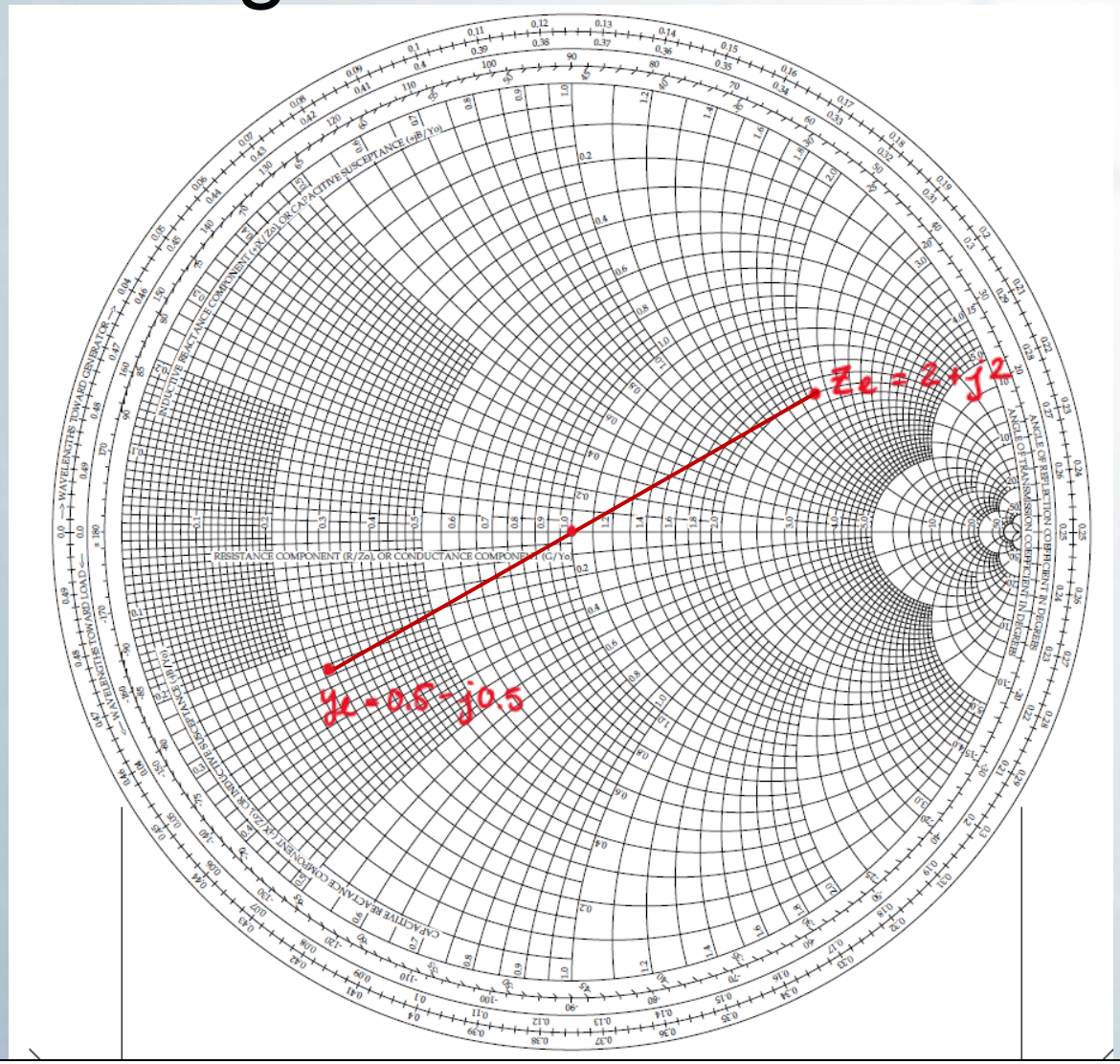
- Load of $Z_L = 100 + j100\,\Omega$ connected to a $Z_0 = 50\Omega$ transmission line
- The frequency is $f = 1\ \mathrm{GHz} = 1\times10^9\ \mathrm{Hz}$
- We want to place an **open circuit stub** somewhere on the line to match the load to the source.
- First, we must determine where to place the stub on the t-line connected to the load. Then, we choose the length of the open circuit stub to attach that matches the source to the load.



Single Stub Matching

- Normalized load impedance:

$$z_L = \frac{Z_L}{Z_0} = 2 + j2$$

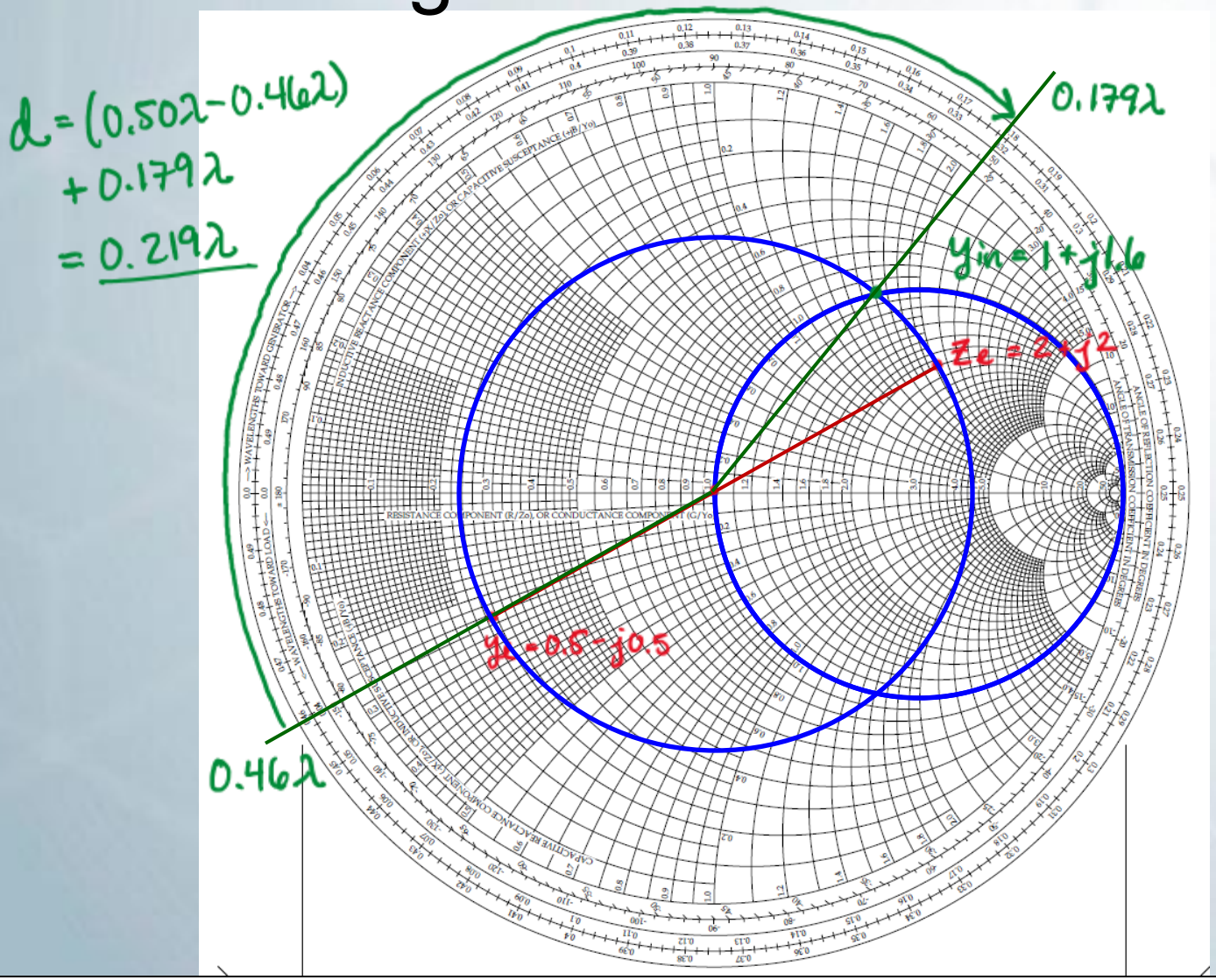


What is the normalized admittance corresponding to the normalized impedance 2+j2?

Single Stub Matching

- How much t-line do we need to transform the admittance $\mathrm{y} = 0.25 - j0.25$ into the admittance $y = 1 + jb$?

Single Stub Matching



How much t-line do we need to transform the admittance $y = 0.25 - j0.25$ into the admittance $y = 1 + jb$?

Single Stub Matching

- How much t-line do we need to transform the admittance $y = 0.25 - j0.25$ into the admittance $y = 1 + jb$?

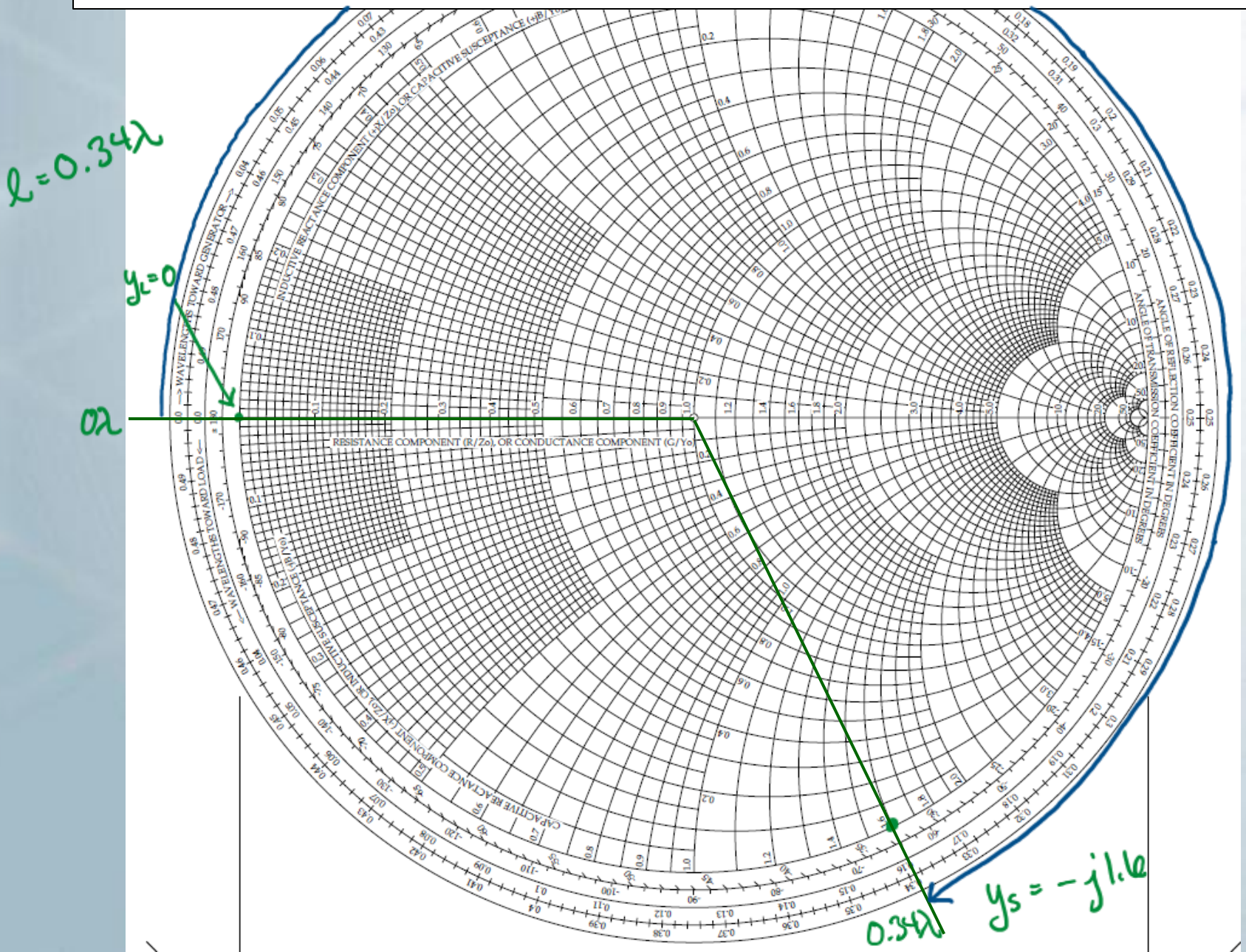
$d = 0.219\lambda$

$y_d = 1 + j1.6$

- This means that we need a stub of admittance -j1.6 to match. If it's an open circuit stub, what length does it need to be?

$\ell = 0.3395\lambda$

This means that we need a stub of admittance -j1.6 to match. If it's an open circuit stub, what length does it need to be?



Single Stub Matching

- We said that this circuit is operating at 1GHz. How do we figure out the actual length 0.34λ of stub? Assume the velocity factor of the t-line is 0.66.

$$\lambda = v/f = \frac{0.66 \cdot 3\times10^{8}\,m/s}{1\times10^{9}\,Hz} \approx 0.198m$$

$$d = 0.219\lambda \approx 4.34\,cm \leftarrow \text{location of stub w.r.t. load}$$

$$l = 0.34\lambda \approx 6.73\,cm \leftarrow \text{stub length}$$

Single Stub Matching

- A single-stub matching network design can lead to 4 possible solutions. In the example just completed, we could have selected:
  - $y_d = 1 + j1.6$ OR $y_d = 1 - j1.6$ (two different locations on matching circle)
  - a short circuit terminated stub OR an open circuit terminated stub

Single Stub Matching

- Which you choose depends upon practical considerations:
  - Can I use open or short circuit terminations in the transmission line I am using?
  - Does it matter if there is a voltage maximum on the line between the stub and the load termination?
  - Is the physical length of line between the stub and the load termination too short/long?

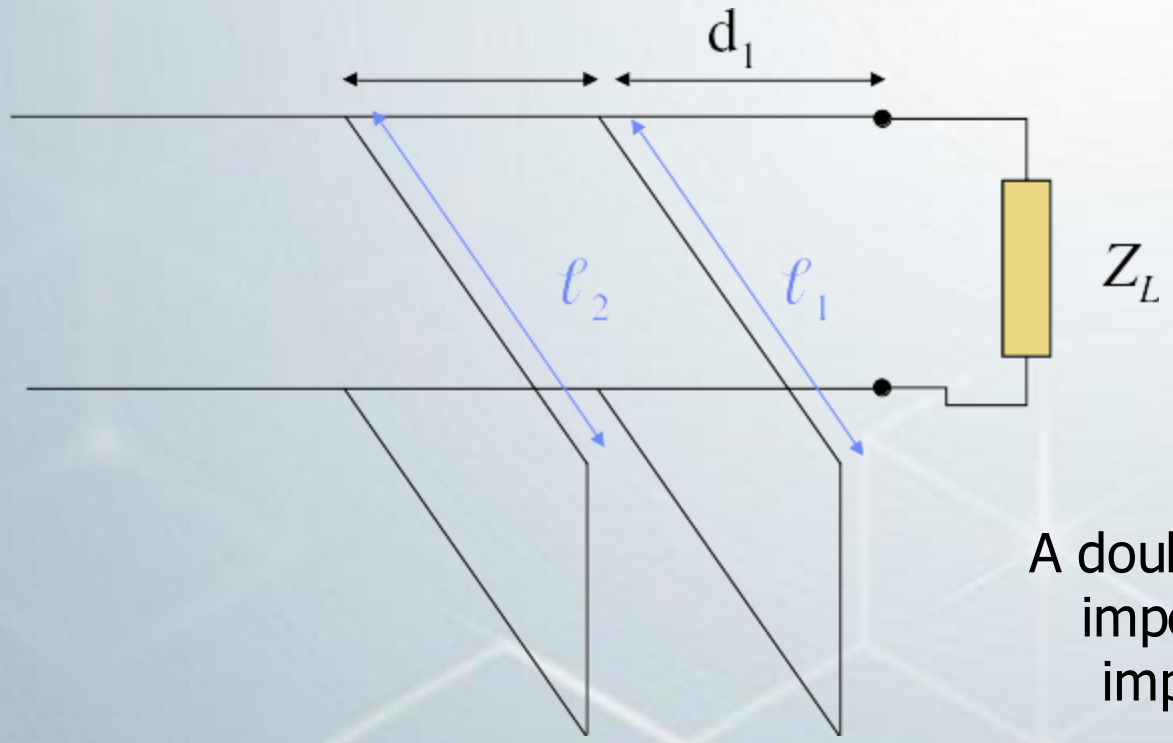
- As an engineer, these are the decisions you must be able to make!

Single Stub Matching

- Single stubs also have a disadvantage: they must be a specific distance from the load in order to work. This isn't always practical.
- Furthermore, what if the frequency or load changes? In practice, this would mean completely disassembling and reassembling the line in order to match it correctly.
- Is there a better way?

Yes: the double stub.

Double Stub Matching



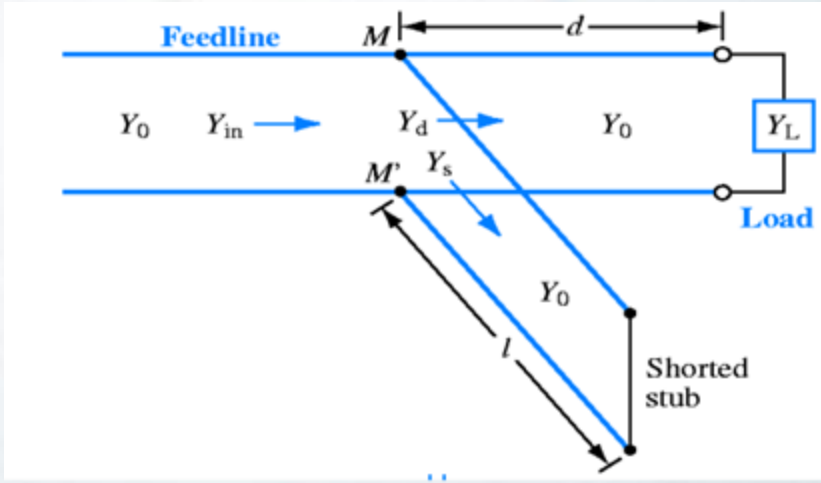
In essence: Use the first stub to transform Z_L into an impedance **that can be matched by the second stub.**

A double stub can match most impedances. To match **all** impedances, you need a **triple stub**.

Project 1: Channel Blocker

Project 1

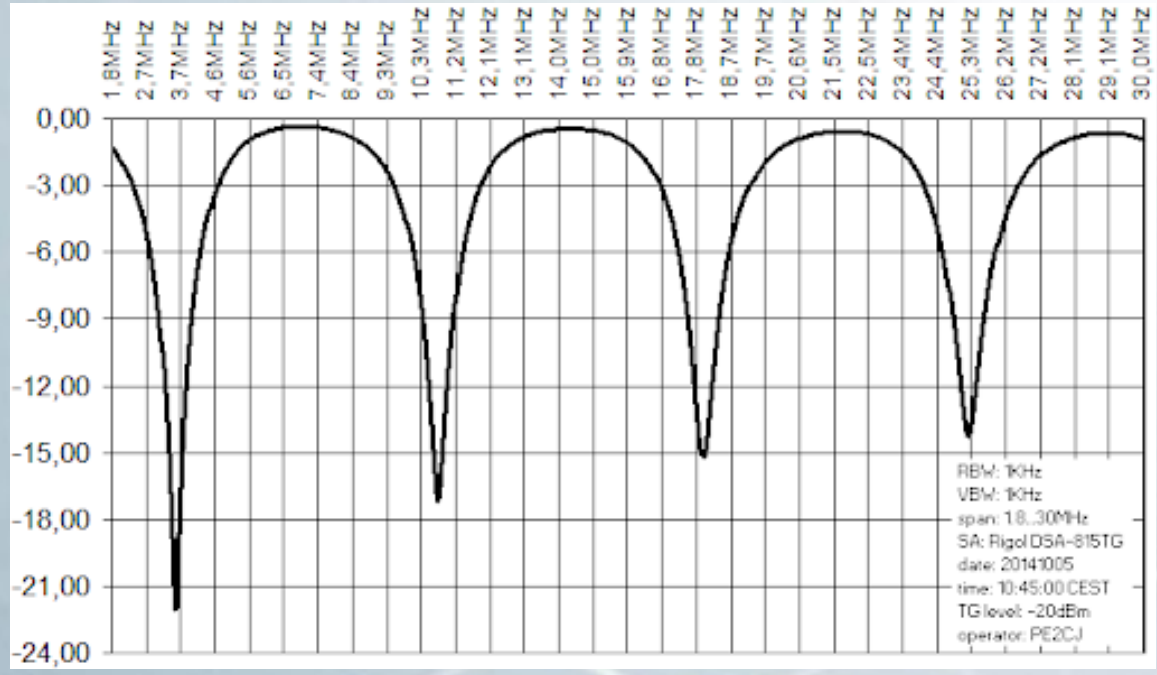
- Stubs can be for more than just matching. They can be used as filters too.
- The quarter-wavelength open circuit stub has zero admittance at the load and infinite admittance at the input. (It's a short circuit.)



- This also works for a half-wavelength short circuit stub.
- This only works at one specific frequency. But there's something useful we can do with this: when we have a band of frequencies on a t-line, we can use the stub to "short" (filter out) one specific frequency.

Project 1

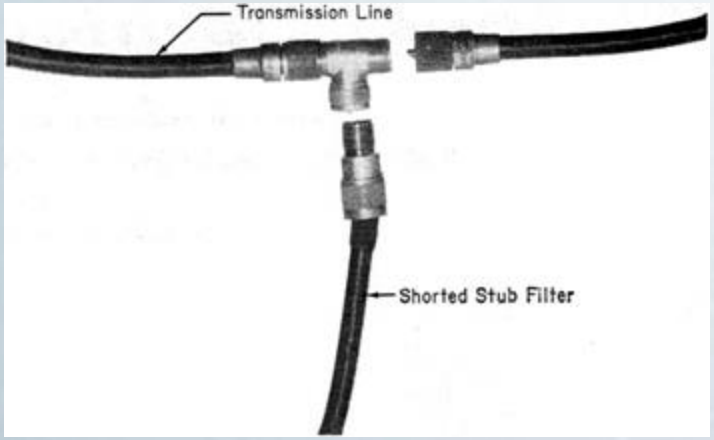
- In practice, this also attenuates frequencies near the blocked frequency (for which the stub has a non-infinite but very large admittance.)



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Project 1

- In Project 1, which starts the Studio Session after Exam 1, you will explore this.
- You will attempt to block TV cable and radio channels using a stub of coaxial cable.



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Wrap-Up

- Instead of a Studio Session on Wednesday, we will have an extra set of office hours in the lab JEC 4107.
- For those who are interested in learning about more advanced stub-matching techniques, there is a supplemental lecture on LMS.